

LUNG CANCER DETECTION USING CNN AND XCEPTION TRANSFER LEARNING

ABSTRACT

Lung cancer is one of the leading causes of cancer-related deaths worldwide. Early and accurate diagnosis is critical for improving patient survival rates. However, manual analysis of chest CT scan images by radiologists is time-consuming and may be subject to human error and variability. With the increasing availability of medical imaging data, there is a growing need for automated systems that can assist in the detection and classification of lung cancer.

This project proposes a Deep Learning-based Lung Cancer Detection System using Convolutional Neural Networks (CNN) with Xception transfer learning. The system classifies CT scan images into three categories: Benign Cases, Malignant Cases, and Normal Cases using the IQ-OTHNCCD Lung Cancer Dataset. Image preprocessing and data augmentation techniques such as rotation, zooming, and horizontal flipping are applied to improve model generalization and classification performance. The pre-trained Xception model is used as the feature extraction backbone, while custom classification layers are added for multi-class prediction. The model is trained and optimized using transfer learning techniques to achieve accurate classification results.

A Streamlit-based web application is developed to provide an interactive interface for uploading CT scan images and obtaining prediction results along with confidence scores. By integrating transfer learning with medical image classification, the system enhances diagnostic support capabilities, reduces manual workload, and demonstrates the potential of artificial intelligence in medical image analysis and clinical decision support systems.

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